

41. Let $f: [-1, 1] \rightarrow \mathbb{R}$ be defined as $f(x) = ax^2 + bx + c$ for all $x \in [-1, 1]$, where $a, b, c \in \mathbb{R}$ such that $f(-1) = 2$, $f'(1) = 1$ and for $x \in (-1, 1)$ the maximum value of $f'(x)$ is $\frac{1}{2}$. If $f(x) \leq \alpha$, $x \in [-1, 1]$, then the least value of α is equal to _____.

माना $f: [-1, 1] \rightarrow \mathbb{R}$, $f(x) = ax^2 + bx + c \forall x \in [-1, 1]$, $a, b, c \in \mathbb{R}$ द्वारा परिभाषित है, जबकि $f(-1) = 2$, $f'(1) = 1$ हैं तथा $x \in (-1, 1)$ के लिए $f'(x)$ का अधिकतम मान $\frac{1}{2}$ है। यदि $f(x) \leq \alpha$, $x \in [-1, 1]$ है, तो α का निम्नतम मान है _____।

Ans. 5

Sol. $f(x) = ax^2 + bx + c$

$$f(-1) = a - b + c = 2 \quad \dots\dots\dots(1)$$

$$f'(x) = 2ax + b$$

$$f'(1) = 2a + b = 1 \quad \dots\dots\dots(2)$$

$$f''(x) = 2a$$

$$(f''(x))_{\text{Max}} = 2a = \frac{1}{2}$$

$$a = \frac{1}{4}$$

$$\text{by (2) } b = 1 - 2a$$

$$b = 1 - \frac{1}{2}$$

$$b = \frac{1}{2}$$

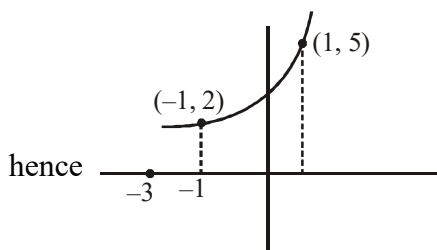
$$\text{by (1) } c = 2 + b - a$$

$$c = 2 + \frac{1}{2} - \frac{1}{4}$$

$$c = 2 + \frac{1}{4}$$

$$c = \frac{8+6-1}{4} = \frac{13}{4}$$

$$f(x) = \frac{1}{4}x^2 + \frac{3}{2}x + \frac{13}{4}$$



$$f(x) \leq \alpha$$

$$\alpha_{\min} = 5$$

42. Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval $(a, a + 1)$. Then, $|a|$ is equal to _____.

माना a एक पूर्णांक है जिसके लिए बहुपद $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ के सभी वास्तविक मूल अन्तराल $(a, a + 1)$ में हैं। तो $|a|$ बराबर है _____।

Ans. 2

Sol. Let $P(x) = 2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$

$$P'(x) = 10x^4 + 20x^3 + 30x^2 + 20x + 10$$

$$P'(x) = 10x^2 \left[x^2 + 2x + 3 + \frac{2}{x} + \frac{1}{x^2} \right]$$

$$= 10x^2 \left[x^2 + \frac{1}{x^2} + 2 \left(x + \frac{1}{x} \right) + 1 \right]$$

$$= 10x^2 \left[\left(x + \frac{1}{x} \right)^2 + 2 \left(x + \frac{1}{x} \right) + 1 \right]$$

$$= 10x^2 \left[x + \frac{1}{x} + 1 \right]^2$$

$$P'(x) = 10 [x^2 + x + 1]^2 > 0$$

$P(x)$ is an odd degree increasing polynomial function

$\Rightarrow$ only one real root

$$P(0) = 10$$

$$P(-1) = -2 + 5 - 10 + 10 - 10 + 10 = 3$$

$$P(-2) = -64 + 80 - 80 + 40 - 20 + 10 = -34$$

$$P(-1) P(-2) < 0$$

Using Intermediate value theorem

$\Rightarrow P(x) = 0$ has at least one root in the $(-2, -1)$

Since $P(x) = 0$ has exactly one real root

$\therefore$ all real roots lie in the interval $(-2, -1)$

$$\therefore a = -2 \quad |a| = 2.$$

43. If the lines $x + y = a$ and $x - y = b$ touch the curve $y = x^2 - 3x + 2$ at the points where the curve intersects the x -axis, then is $\frac{a}{b}$ equal to _____.

यदि रेखाएँ $x + y = a$ तथा $x - y = b$, वक्र $y = x^2 - 3x + 2$ को उन बिन्दुओं पर स्पर्श करती हैं जहाँ यह वक्र x -अक्ष को काटता

है, तो $\frac{a}{b}$ बराबर है _____ ।

Ans. 0.5

Sol. Curve touch x -axis at axis $y = 0$

$$x^2 - 3x + 2 = 0$$

$$x = 1, x = 2$$

$$P(1, 0) Q(2, 0)$$

$$dy = 2x - 3$$

$$\left. \frac{dy}{dx} \right|_{(1,0)} = -1$$

equation of tangent at P

$$y - 0 = -1(x - 1)$$

$$x + y = 1$$

$$\left. \frac{dy}{dx} \right|_{(2,0)} = 1$$

equation of tangent at Q

$$y - 0 = x - 2$$

$$x - y = 2$$

44. If $f(x) = x^2 + g'(1)x + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$, then the value of $f(4) - g(4)$ is equal to _____.

यदि $f(x) = x^2 + g'(1)x + g''(2)$ एवं $g(x) = f(1)x^2 + xf'(x) + f''(x)$ है, तो $f(4) - g(4)$ का मान है।

Ans. 14

Sol. $f(x) = x^2 + xg'(1) + g''(2) \quad \dots(1)$

$$\therefore f'(x) = 2x + g'(1)$$

$$f''(x) = 2$$

$$\therefore g(x) = f(1)x^2 + x(2x + g'(1)) + 2 = f(1)x^2 + 2x^2 + xg'(1) + 2$$

$$g'(x) = 2f(1)x + 4x + g'(1) \quad \dots(2)$$

$$g''(x) = 2(1) + 4 \quad \dots(3)$$

from (2) put $x = 1$

$$g'(1) + 2f(1) + 4 + g'(1)$$

$$\Rightarrow 2f(1) + 4 = 0$$

$$\Rightarrow f(1) = -2 \Rightarrow g''(2) = -4 + 4 = 0 \quad \text{from (3)}$$

from (1)

$$f(1) = 1 + g'(1) + 0$$

$$\Rightarrow g'(1) = -2 - 1 = -3$$

$$\therefore f(x) = 2x - 3$$

$$f(x) = x^2 - 3x + c$$

$$\text{put } x = 1, f(1) = -2 \quad \therefore c = 0$$

$$\therefore f(x) = x^2 - 3x$$

$$\therefore g(x) = -3x + 2$$

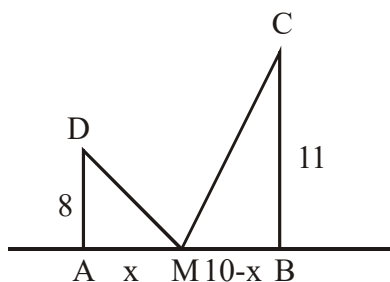
$$\Rightarrow f(4) - g(4) = 4 - (-10) = 14$$

45. Let AD and BC be two vertical poles at A and B respectively on a horizontal ground. If $AD = 8$ m, $BC = 11$ m and $AB = 10$ m; then the distance (in meters) of a point M on AB from the point A such that $MD^2 + MC^2$ is minimum is _____.

माना AD तथा BC क्षैतिज समतल भूमी पर क्रमशः A तथा B पर सीधे खड़े खम्भे हैं। यदि $AD = 8$ मीटर, $BC = 11$ मीटर तथा $AB = 10$ मीटर ह, तो AB पर स्थित एक बिंदु A से वह दूरी (मीटरों में) जिसके लिए $MD^2 + MC^2$ का मान न्यूनतम है, है _____।

Ans 5

Sol.



$$MD^2 + MC^2 = x^2 + 64 + (10 - x)^2 + 121$$

$$MD^2 + MC^2 = 2x^2 - 20x + 285$$

$$MD^2 + MC^2 = 2(x - 5)^2 + 235$$

$$(MD^2 + MC^2)_{\min} \text{ when } x = 5$$